

Unit 02 (Descriptive Statistics): Practice Problems

Instructor: Md Mutasim Billah, PhD

Required Practice Problems

Complete these required practice problems throughout the unit as we cover the corresponding topics in class. These problems reinforce the concepts discussed in the lecture notes, handouts, and course resources. Before attempting the homework assignments and preparing for exams, make sure you have completed and reviewed all practice problems from the unit.

2.1

Use this data set to draw a stem-and-leaf plot:

0.9	5.8	6.6	4.3	8.8	6.4	3.2	6.1	2.0	5.3	7.2	8.8	8.6	8.3	1.5	7.7	4.1
5.6	8.1	5.4	7.0	8.2	7.4	7.7	7.0	8.0	7.6	6.9	3.6	6.7	4.6	8.3	8.4	

2.2

Twelve left-handed pitchers were randomly selected and their heights were rounded to the nearest tenth of an inch. Use the data to draw a stem-and-leaf plot:

71.2, 71.7, 72.6, 72.6, 73.0, 73.5, 74.1, 74.9, 75.2, 75.3, 76.4, 76.6

2.3

Refer to Practice Problem 2.1.

Describe the shape of the stem-and-leaf plot.

2.4

Refer to Practice Problem 2.2.

Describe the shape of the stem-and-leaf plot.

2.5

Refer to Practice Problem 1.1.

The historian visits the 15 bells he randomly selected and measures their loudness in decibels (dB) when rung.

The data are:

90.2	98.7	92.4	94.7	97.6	98.1	98.3	93.8
99.9	95.9	96.2	94.3	99.3	99.5	97.9	

Draw a histogram using the classes:

$[90, 92)$, $[92, 94)$, $[94, 96)$, $\dots$

Describe the shape of the histogram.

2.6

Use this data to draw a histogram and have the classes be $[0, 5)$, $[5, 10)$, $[10, 15)$, $\dots$ etc. Describe the shape of the histogram.

0.107 0.741 0.849 1.590 2.686 3.432 4.114 4.253 5.107 5.754 7.423 9.031
9.315 9.413 10.108 12.041 12.394 13.085 13.997 15.108 19.034 22.222

Draw the histogram and describe its shape.

2.7

Three data sets are described below. Choose exactly one of the following descriptions for each data set:

- The mean is less than the median.
- The mean is greater than the median.
- The mean and median are approximately equal.

a)

The owner of a car dealership is looking at a list of the mileages on all the cars available on the lot. Almost all of the cars are brand new, so almost all of the mileages are close to zero. However, just a few are used cars with mileages greater than 80,000 miles.

b)

A doctor is looking at a histogram from a data set of blood pressure measurements. The histogram is symmetric and bell-shaped, with most observations (tallest rectangles) in the middle.

c)

A professor is looking at a large data set of exam scores. Almost all of the scores are between 70–90 points. However, a small handful of students had extremely low scores in the neighborhood of 10–20 points.

2.8

Refer to Practice Problem 2.2.

Find the sample mean and median among the heights.

2.9

Refer to Practice Problem 2.5.

Find the sample mean and median among the loudness measurements.

2.10

In a data set with $n = 13,642$ values, where would you find the median? What if $n = 173,209$?

2.11

Find the sample variance of the data set

$$\{0, 3, 3, 9, 10, 13, 18\}$$

using both the “regular” formula and “convenient” formula.

Make a table and show your work. You should get the same answer with both formulas. Finally, find the standard deviation.

2.12

Find the sample variance of the data set

$$\{5, 8, 9, 10, 17, 21, 24, 30\}$$

using both the “regular” formula and “convenient” formula.

Make a table and show your work. You should get the same answer with both formulas. Finally, find the standard deviation.

2.13

At a large high school exactly 100 juniors and 100 seniors are assembled in the auditorium. If you were to ask the students what grade they are in, the juniors would respond with “11” and the seniors would respond with “12”.

What is the sample mean of this data set? What is the approximate value of the standard deviation?

2.14

The members of a book club calculate the sample mean and standard deviation among their ages.

$$\bar{X} = 34.1 \quad \text{and} \quad s = 7.2$$

What will be the sample mean and standard deviation among their ages exactly 5 years from today?

2.15

Each slot machine in a Las Vegas casino contains a certain number of quarters. Suppose the sample mean and standard deviation among the numbers of quarters are

$$\bar{X} = 512.4 \quad \text{and} \quad s = 83.5$$

What is the sample mean and standard deviation among the dollar amounts of these quarters?

$$(1 \text{ quarter} = 0.25 \text{ dollars})$$

2.16

Create your own data sets, each with a sample size of $n = 5$, that satisfy the following descriptions. If no such data set could exist, write “impossible.”

- a) Both the mean and median equal ten, and the standard deviation is positive.
- b) The standard deviation is negative.
- c) Four of the five values are less than the mean.
- d) The mean is negative, but the median is positive.

2.17

Refer to Practice Problem 2.2.

Find the five number summary for the heights.

2.18

Refer to Practice Problem 2.5.

Find the five number summary for the loudness measurements. Remember to order the data first.

2.19

In a data set with $n = 13,642$ values, where would you find the three quartiles?

What if $n = 173,209$?

2.20

Sixteen cigarette smokers made a New Year’s resolution to quit smoking. Ten weeks later they were asked how many cigarettes they smoked yesterday.

The responses were:

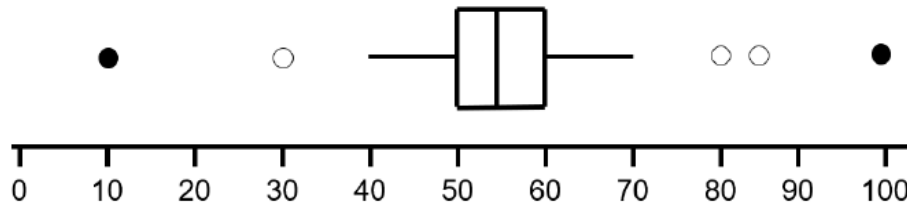
$\{0, 0, 1, 2, 3, 3, 6, 7, 9, 10, 10, 10, 11, 13, 19, 26\}$

Find the five number summary, IQR, the locations of the inner and outer fences, and then use these values to draw a boxplot.

2.21

Examine the following boxplot.

Find the value of the range, IQR, and the location of the fences.



2.22

Solve the following exercises from the “Chapter 6 Exercises” section of the textbook (*Applied Statistics and Probability for Engineers*, 7th edition, by Montgomery and Runger).

- 6.11–6.15, 6.17, 6.22, 6.48

[To find these exercises in the digital copy of the textbook (**Textbooks > Files > Canvas**), go to the xvi Contents page and click the “EXERCISES” tab located at the lower-right corner of the page. This will open the complete list of chapter exercises.]

Supplemental Practice Problems

The problems below provide additional practice opportunities for students who would like more practice with the concepts covered in this unit.

Chapter 6 (*Applied Statistics and Probability for Engineers*, 7th edition, by Montgomery and Runger)

- 6.1.8, 6.2.5 (Only construct the sem-and-leaf display for this problem), 6.2.7, 6.4.3, 6.4.4